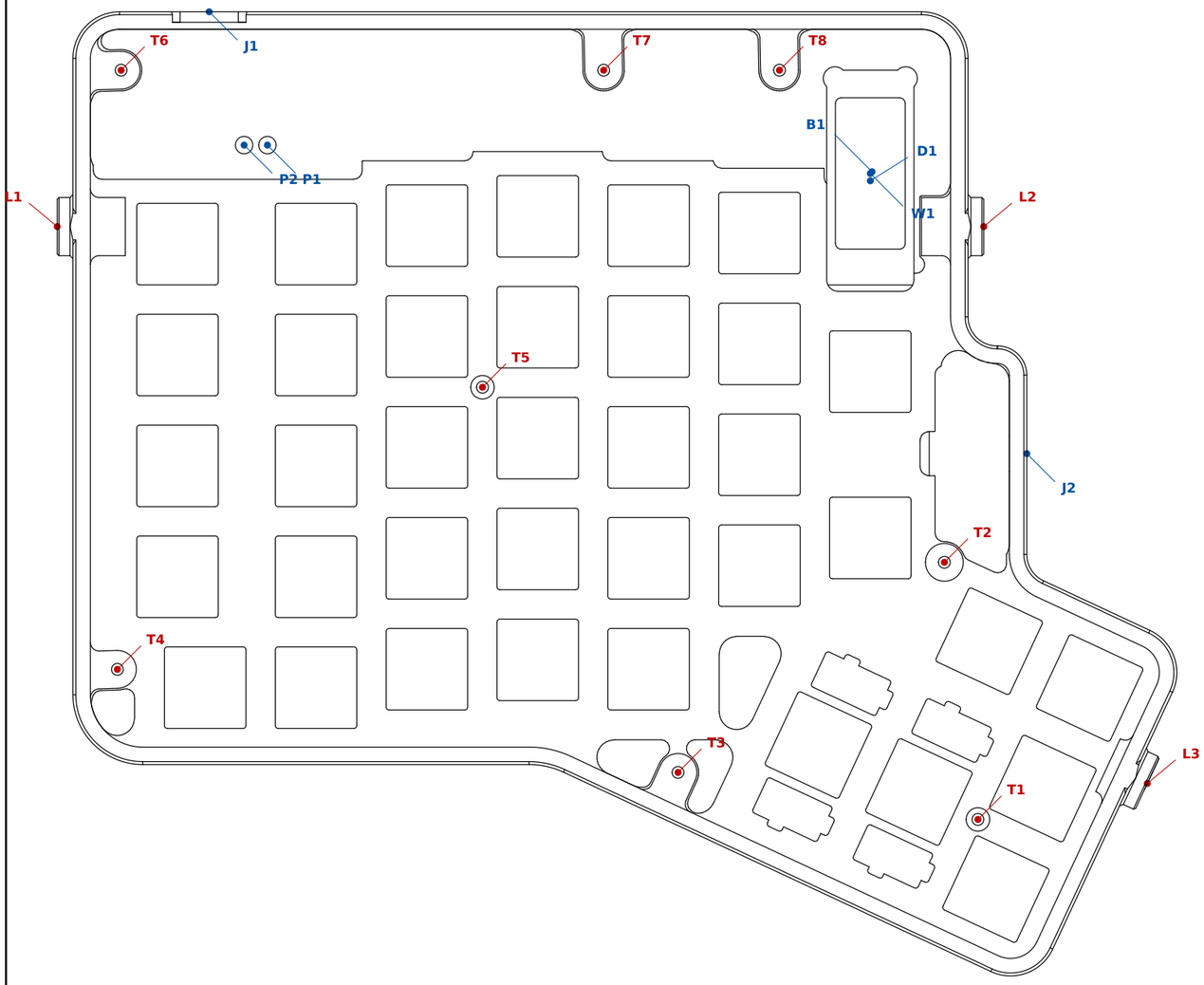
## VIEW FROM BELOW (−Z), parting face up — scale 1:1




| ID | Thread | Hand | Cls | Tap drill | Drill depth | Min full thr. | Screw | X | Y | Z | Note |
|---|---|---|---|---|---|---|---|---|---|---|---|
| T1 | M2.5x0.45 | RH | 6H | Ø2.05 | 3.6 | 2.3 | M2.5x10 CSK | -76.10 | -65.84 | 8.6 | BLIND under key well, 0.78 skin: no break-through |
| T2 | M2.5x0.45 | RH | 6H | Ø2.05 | 6.0 | 4.5 | M2.5x12 CSK | -70.34 | -21.67 | 8.6 | blind |
| T3 | M2.5x0.45 | RH | 6H | Ø2.05 | 6.0 | 4.5 | M2.5x12 CSK | -24.56 | -57.76 | 8.6 | blind |
| T4 | M2.5x0.45 | RH | 6H | Ø2.05 | 6.0 | 4.5 | M2.5x12 CSK | 71.78 | -40.06 | 8.6 | blind |
| T5 | M2.5x0.45 | RH | 6H | Ø2.05 | 3.6 | 2.3 | M2.5x10 CSK | 9.04 | 8.41 | 8.6 | BLIND under key well, 0.78 skin: no break-through |
| T6 | M2.5x0.45 | RH | 6H | Ø2.05 | 6.0 | 4.5 | M2.5x12 CSK | 71.14 | 62.87 | 8.6 | blind |
| T7 | M2.5x0.45 | RH | 6H | Ø2.05 | 6.0 | 4.5 | M2.5x12 CSK | -11.79 | 62.87 | 8.6 | blind |
| T8 | M2.5x0.45 | RH | 6H | Ø2.05 | 6.0 | 4.5 | M2.5x12 CSK | -42.00 | 62.87 | 8.6 | blind |
| L1 | M3x0.5 | RH | 6H | Ø2.5 | 5.5 | 4.0 | M3 leg thumbscrew | 82.12 | 36.00 | 12.0 | blind, in the flat leg boss, see sheet 2 |
| L2 | M3x0.5 | RH | 6H | Ø2.5 | 7.5 | 6.0 | M3 leg thumbscrew | -77.10 | 36.00 | 12.0 | blind, in the flat leg boss, see sheet 2 |
| L3 | M3x0.5 | RH | 6H | Ø2.5 | 5.5 | 4.0 | M3 leg thumbscrew | -105.21 | -59.66 | 12.0 | blind, in the flat leg boss, see sheet 2 |

X/Y/Z = hole start point in STEP coordinates. T holes: drilled +Z from the PCB face of the boss. L holes: drilled into the part, normal to the wall, from the boss face.
Drill depth = full-diameter depth; the 118° drill point is extra (modelled). Every hole is BLIND and keeps at least D/2 unthreaded at its bottom (JLCCNC threaded-hole guideline).

| ID | Feature (no thread) | X | Y |
|---|---|---|---|
| W1 | display window 12 x 26 R1.0 THRU the bezel skin, 0.5 x 45° chamfer on the top edge | -57.61 | 45.13 |
| D1 | display pocket 15.0 x 38.0, Z 8.6-18.6 (skin 1.0), 11.6 deep from the parting face: R2.0 dog-bones at the 2 back corners (keep), front end open into the key well | -57.61 | 43.87 |
| B1 | raised display bezel 18.42 x 38.90, top Z 19.6 (2.0 above the top face), top edges 0.5 x 45°, R2.0 free corners; flush with the key-well walls on its well sides | -57.90 | 45.43 |
| J1 | USB-C notch 9.4 x 3.6 R1.6 thru back wall at Z 10.2; outside recess 13 x 8 x 1.8 deep | 56.00 | 72.90 |
| J2 | TRRS hole Ø6.5 thru inner bump wall, axis Z 11.1 | -84.50 | -3.00 |
| P1 | Ø3.0 THRU top skin (SW_BOOT pin-hole) | 46.00 | 50.00 |
| P2 | Ø3.0 THRU top skin (SW_RST pin-hole) | 50.00 | 50.00 |

### GENERAL NOTES

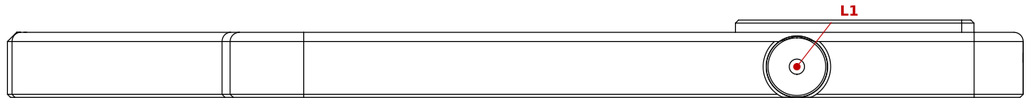
1. Geometry per worxboard_right_top.step. Dimensions not shown are taken from the STEP model.
2. Material: Aluminium 6061-T6 (7075-T6 acceptable).
3. THREADS: 8x M2.5x0.45-6H and 3x M3x0.5-6H, ISO metric coarse, ALL RIGHT-HAND (clockwise to tighten), all blind. Holes are modelled at tap-drill diameter in the STEP; tap them per the table to at least the minimum full-thread depth. The screw column names the screw each hole takes; its tip stops at least 0.3 short of the end of the full thread.
4. The left-hand part is a mirror image of this part. Its threads are also RIGHT-HAND.
5. Holes marked BLIND under key well: drill 3.6 deep (full diameter); 0.78 mm of material remains between the drill point and the cosmetic key-well floor. Must not break through or leave witness marks on the top surface.
6. Switch cutouts (37x): 14.00 +0.05/-0 square, corner R0.5 max, plus 4 stabiliser cutouts. Plate thickness 1.50 ±0.05. Plate flatness 0.2 over the part.
7. Internal corners are modelled at JLCCNC's minimum radius for their depth (R = depth/10 + 0.5; past 10 mm the Min R of the tool that reaches): key well R1.0 (4.0 deep), relief pockets and wall fillets R1.5 (to 7.6 deep from the parting face), pocket D1 R2.0 (11.6 deep, dog-bones at its back corners). Only the switch and stabiliser cutouts are R0.5 (Ø1 tool, through the 1.5 plate at the bottom of the open key well).
8. Display bezel B1 stands 2.0 mm above the top face (part height 12.6). Pocket D1 runs from the parting side up to 1.0 mm under the bezel top; window W1 opens it. USB-C notch J1 and TRRS hole J2 are side operations (see table).
9. Leg bosses L1-L3: plain Ø10 bosses with a FLAT face (no teeth), see sheet 2.
10. Cosmetic surfaces: top face, display bezel, key well, outer walls, leg bosses: no tool marks, scratches or clamp marks. Inside (cavity) surfaces are non-cosmetic.
11. Break sharp edges 0.1-0.2 mm, except switch-cutout edges (0.1 max).
12. Finish: Fine bead blast + anodize Type II, colour per order. Mask threads during anodizing, or re-tap after anodizing.

## worxboard CNC aluminium case

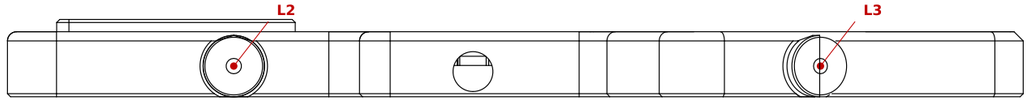
worxboard_right_top — https://github.com/w0rxbend/worxboard

| | |
|---|---|
| Drawing no.: WXB-TOP-R | Sheet 1 of 2 |
| Material: Aluminium 6061-T6 (7075-T6 acceptable)<br>Finish: Fine bead blast + anodize Type II, colour per order | Scale 1:1<br>Units: mm |
| Revision A4 Date 2026-11-07<br>General tolerance ISO 2768-mK. Threads ISO metric, RIGHT-HAND, 6H.<br>The STEP file governs geometry; this drawing governs threads, tolerances, finish. | Do not scale drawing |

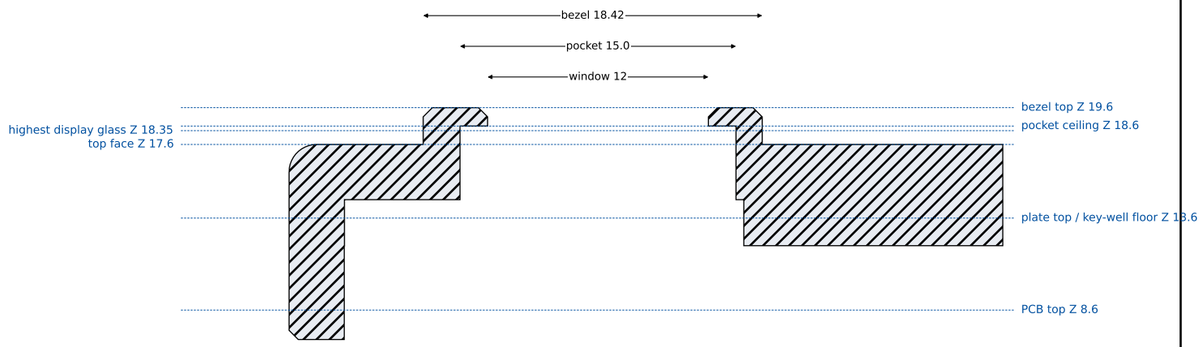
## VIEW FROM +X — scale 1:1




## VIEW FROM −X — scale 1:1

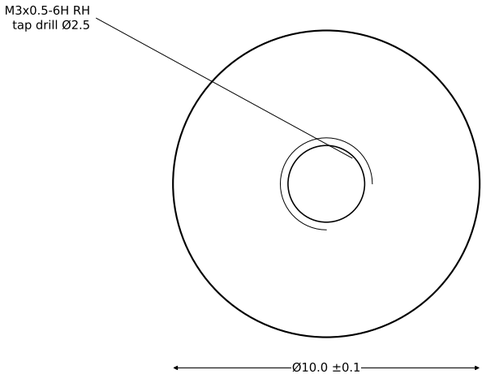



## DETAIL C — SECTION THROUGH THE DISPLAY BEZEL at the window centre (X-Z, real section), scale 3:1

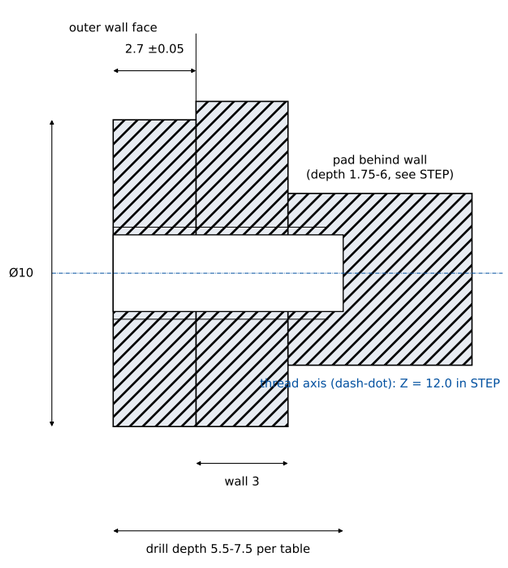



bezel skin 1.0; top edges 0.5 x 45°; window edge 0.5 x 45°. The bezel side wall on the key-well side is flush with the well wall.

## DETAIL A — LEG BOSS L1-L3, face view, scale 5:1 (FLAT FACE, NO TEETH)




## SECTION B-B through boss axis, scale 5:1 (schematic)




| Leg boss specification | |
|---|---|
| Boss | Ø10.0 ±0.1, 2.7 ±0.05 proud of the outer wall face |
| Bearing face | FLAT, no teeth. Flatness 0.05, perpendicular to the thread axis within 0.05. Ra 1.6 max |
| Function | the leg hub is clamped against this face by the M3 spring thumbscrew (friction) |
| Thread | M3x0.5-6H RIGHT-HAND, blind, concentric with boss within Ø0.05; tap drill Ø2.5; depths per the table below |
| Quantity | 3 bosses per part (L1-L3), identical |
| Machining hint | side op (4th axis, right-angle head or a setup per wall): face the boss, profile Ø10, drill and tap M3 |
| Mating part | custom leg worxboard_leg (drawing WXB-LEG, flat hub face) + M3 thumbscrew |

| Boss | X face | Y face | Z | Thread direction (into part) | Drill depth | Min full thr. | Screw reach max |
|---|---|---|---|---|---|---|---|
| L1 | 82.12 | 36.00 | 12.0 | into part (-1.000, -0.000, 0) | 5.5 | 4.0 | 3.7 |
| L2 | -77.10 | 36.00 | 12.0 | into part (+1.000, -0.000, 0) | 7.5 | 6.0 | 5.7 |
| L3 | -105.21 | -59.66 | 12.0 | into part (+0.906, +0.423, 0) | 5.5 | 4.0 | 3.7 |

X/Y/Z: centre of the boss face (tooth tip plane) in STEP coordinates. Depths from the boss face; the drill point is extra. >= 1.5 (D/2) stays unthreaded.

# worxboard CNC aluminium case

worxboard_right_top — https://github.com/w0rxbend/worxboard

| | |
|---|---|
| Drawing no.: WXB-TOP-R | Sheet 2 of 2 |
| Material: Aluminium 6061-T6 (7075-T6 acceptable) | Scale 1:1, 3:1, 5:1 |
| Finish: Fine bead blast + anodize Type II, colour per order | Units: mm |
| Revision A4 Date 2026-11-07 | Do not scale drawing |
| General tolerance ISO 2768-mK. Threads ISO metric, RIGHT-HAND, 6H. | |
| The STEP file governs geometry; this drawing governs threads, tolerances, finish. | |